Disaster Impact

Unit -III

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Disaster Impacts- Disaster impacts (environmental, physical, social, ecological, economic, political, etc.); health, psycho-social issues; demographic aspects (gender, age, special needs); hazard locations; global and national disaster trends; climate change and urban disasters, capacity building – concepts, assessment –structural & non-structural measures, legislative support.

Disaster impact:

- INTRODUCTION
- The rising Costs and Frequency of Natural Disasters in India, the risks related to natural hazards have increased alarmingly in recent years, with the number of recorded natural disasters having increased exponentially since 1947.
- Since the year 2000 more than 300 disasters including earthquake, drought, epidemics, floods, storms, landslides and extreme temperatures have occurred in India, leading to 76,031 casualties till 2017, according to the International Disaster Database(IDD).
- More than 1 billion people have suffered owing to disasters.
- The estimated damage in these 17 years amounts to USD 63.6 billion(Rs 4,06,035 crore), based on the data.

THE SHORT-TERM AND LONG-TERM IMPACT OF NATURAL DISASTERS

Short Term Impact:

- Disasters may destroy various assets of businesses varying from buildings and equipment to even their workforce.
- When businesses suffer, they tend to impact market trade and buying power of consumers as well.
- After getting hit by a natural disaster, it takes months for a business to return to its normal functioning as they have to wait for insurance claims and for its cash flow to return to normal.
- Small businesses suffer from severe negative impact of natural disasters as compared to large businesses since they do not have large capital resources to fund themselves.
- They take longer to recover and even lose customer trust.
- This is a major reason behind the shutting off of small businesses just within a short span of time after being hit by a disaster leading to a loss in GDP of India.
- For example: In Chennai, where waterlogging affected many areas for a long time, businesses began staggered operations after 1–2 weeks.

- SMEs were without power, water, and other basic services for 10–15 days, impacting business continuity.
- A few SMEs that had opted for flood insurance received less than the claimed amount after months of delay.
- As the losses from floods were unexpected, businesses took a long time to recover.
- Some of these businesses could not even repay the loans which they took from financial institutions.
- This increased the stress of some of these businesses and as a repercussion they sold their assets and even closed operations.

Long Term Impact

- In the long run, economic hardship is caused by loss of livestock, crops, industries, etc.
- The tourism industry is the one which is impacted by the negative effects of disasters in the long run as tourists tend to avoid disaster-prone areas.
- For example: Cyclone Gaja affected more than 14 districts in Kerala.
- Kerala- a place of beauty where there are 418 hotels and 12,672 rooms and 30% of income for the state is from tourism .
- After the cyclone, the economy of the state had fallen as many bookings were cancelled and the tourism-related industries were closed for months.
- Ernakulam district, Idukki district (Munnar, Thekkady) were the most affected.
- It is the social and economic factors that seem to aid and hinder growth in the long run for developing countries such as India.
- However, studies show that such disasters provide nations with opportunities to update their capital stock and adopt new technologies.

- It was expected that such natural disasters would have a severe impact on the GDP per capita growth of India but no such result was found in the short and long term.
- Electric power consumption played a major role in stimulating growth in the short run while having a high private credit to GDP ratio seemed to hinder economic growth.
- The fact that India is not dependent on international debt for its disaster management unlike other lower income countries provided a relief to the Indian economy as well.

Impacts of Disasters on Environment

- Disasters are not random and do not occur by accident.
- They are the convergence of hazards and vulnerable conditions.
- Disasters not only reveal underlying social, economic, political and environmental problems, but unfortunately contribute to worsening them.
- Such events pose serious challenges to development, as they erode hard-earned gains in terms of political, social and educational progress, as well as infrastructure and technological development.
- Disasters create substantial environmental degradation and ecological imbalance, hinder socioeconomic development and retard the process of improving the quality of life of the people.
- The interaction of disasters and environment has both short-term and long-term effects.
- These interaction and interdependencies work in a complicated way, affecting people, ecosystem and bio-diversity.

Environmental vulnerability

- Reduced access to clean air, safe water and sanitation and inappropriate forms of waste management, especially in heavily populated and urban environments, can aggravate socio economic vulnerability.
- Poorer environmental conditions such as diminished biodiversity, soil degradation or growing food scarcity can easily threaten food security for people dependent on the products of land, forests, pastures and marine environment for their livelihoods.
- As natural resources become scarce, the range of options available to communities become more limited, reducing the availability of coping solutions and reducing local resilience to hazards or capacity to recover from disasters.
- Over a period of time, environmental factors can further increase vulnerability by creating new and undesirable patterns of social discord (lack of agree), economic destitution and eventually forced migration of entire communities.

Physical effects

- Physical destruction of houses and other built structures by the force of the wind, ocean waves, floods, mud slides, avalanches, lava flows and earthquakes
- Sinking of ships by the combined forces of wind and wave, sometimes in combination with ocean currents and/or sea ice
- Inundation of coastal areas, roads, houses, river deltas etc, by storm surges and wind-driven waves, sometimes exacerbated by tidal effects and river flooding
- Inundation of farms, houses, roads, bridges and towns by local heavy rain (flash floods) and riverine floods
- Freezing of crops, stock, water pipes, transport systems etc by extreme cold
- Burning of buildings and forests by fires caused by natural (eg. lightning) or human (eg. motor vehicle) ignition in circumstances of extreme heat and wind, including ignition by electrical faults and gas leaks resulting from earthquake damage; and or by burning ash from volcanic eruptions
- Desertification by the combined effects of recurring drought, overgrazing and other unsustainable land management practices
- Loss of human life from the impacts of all of the above, singly or in combination

Social

1. Displacement and Migration

- **Rural to Urban Migration**: Many people from rural areas are forced to move to urban centers after disasters, creating new patterns of internal migration. This can strain urban resources and infrastructure and lead to the growth of informal settlements (slums).
- **Inter-State Migration**: Large-scale displacement, particularly in flood-prone areas like Bihar, Assam, and Odisha, can lead to significant movements of people between states, altering the demographic composition of both the origin and destination locations.
- **Long-Term Migration**: People displaced due to disasters often find it difficult to return to their previous homes due to loss of livelihood, social connections, and sometimes due to the destruction of property. As a result, many become permanent migrants.

2. Social Inequality and Vulnerability

- **Disproportionate Impact on Marginalized Groups**: Vulnerable groups such as Dalits, Adivasis (tribal communities), women, children, the elderly, and those living in poverty often bear the brunt of disasters. These groups may face more significant challenges in terms of access to resources, relief efforts, and recovery.

- **Caste and Gender Dynamics**: In many cases, caste and gender-based inequalities are exacerbated in post-disaster scenarios. For example, in the aftermath of the 2004 tsunami, women, particularly from lower-caste communities, faced higher rates of displacement and had fewer resources to recover.
- **Loss of Livelihoods**: The economic impact of disasters can drive further inequality. For instance, farmers whose crops are destroyed may face long-term food insecurity and poverty, while wealthier people or those with non-agricultural livelihoods might recover more quickly.

3. Community Solidarity vs. Tensions

- **Community Support and Resilience**: In some cases, disasters foster strong bonds within communities. Mutual aid, collective action, and social solidarity can emerge as people work together to rebuild their homes, livelihoods, and infrastructures. This can strengthen social cohesion.
- **Conflict and Tensions**: On the other hand, resource scarcity, unequal distribution of aid, and competition for relief materials can exacerbate social tensions.
- Disputes over land, water, and other resources often become more pronounced post-disaster.
- In communal areas, these tensions can spill over into violence, especially if the disaster amplifies existing social divisions.

4. Political and Institutional Response

- **Government Response and Accountability**: The effectiveness of the state's response to disasters is crucial in shaping social outcomes. Inadequate or delayed responses can lead to public dissatisfaction, loss of trust in authorities, and in some cases, political instability. Disasters often highlight the weaknesses in governance, infrastructure, and policy, leading to demands for greater accountability.
- **Aid Distribution and Corruption**: The distribution of relief and reconstruction aid can be a source of conflict, as disaster relief often becomes politicized. Inequitable distribution, corruption, or the misuse of funds may further deepen existing social cleavages.

5. Psychological and Emotional Impact

- **Mental Health Crisis**: The trauma of losing loved ones, homes, and livelihoods has a significant psychological toll. Mental health support in the aftermath of disasters is often inadequate, especially in rural and underserved areas. This can lead to long-term issues, such as depression, anxiety, and post-traumatic stress disorder (PTSD).
- **Community Rebuilding**: Social ties and traditional community networks play a significant role in rebuilding emotional resilience. However, in the aftermath of disasters, there is often a loss of these networks as people are displaced, leading to emotional and social isolation for some individuals, especially women and children.

6. Cultural Disruption and Resilience

- **Destruction of Cultural Heritage**: Disasters can lead to the destruction of historically and culturally significant sites, monuments, and traditional homes, leading to a loss of identity and heritage. For instance, the 2001 Gujarat earthquake caused significant damage to ancient temples and structures that held cultural value.
- **Revival of Traditional Practices**: On the other hand, disasters sometimes prompt a revival of cultural traditions. In the aftermath of disasters, communities often turn to traditional knowledge and practices for rebuilding. Indigenous knowledge of sustainable construction, farming, and disaster preparedness can become more prominent in such times.

7. Impact on Social Institutions

- **Education Systems**: Disasters often disrupt educational systems, particularly in rural or remote areas. School buildings may be destroyed, and children may face long periods of displacement, which affects their learning and socialization.
- **Healthcare Systems**: Healthcare services are often overwhelmed, and in many rural or underdeveloped areas, there may be a lack of proper medical facilities. Long-term healthcare infrastructure may also suffer, leaving communities vulnerable to other diseases and health crises.

8. Environmental Changes and Social Adaptation

- **Impact on Agriculture and Livelihoods**: In agrarian societies like India, natural disasters, especially floods, droughts, and cyclones, can devastate agricultural productivity, leading to food insecurity. The social fabric of rural communities often revolves around agriculture, and when farming livelihoods are destroyed, it can create a ripple effect on local economies, leading to increased poverty and social unrest.
- **Adaptation to Climate Change**: Recurrent disasters, particularly those related to climate change such as rising sea levels and increased cyclone activity, are prompting social and cultural adaptation. Communities are adopting new coping strategies, such as changing agricultural practices, seeking alternative livelihoods, or even relocating.

Disaster impact on Economy (Economic impacts)

1. Agricultural Losses

- **Crop Destruction**: India is heavily dependent on agriculture, with a large portion of the population relying on farming for their livelihood. Disasters like floods, droughts, cyclones, and unseasonal rains cause extensive damage to crops, leading to direct economic losses.
- For instance, the 2019 floods in Kerala severely impacted rice, banana, and rubber crops, causing a loss of livelihood for thousands of farmers.

- **Reduction in Yield**: Even in non-destroyed areas, disasters often cause a reduction in crop yields due to damage to the soil, inadequate irrigation, or pests and diseases that follow floods or droughts. This results in lower production, higher food prices, and a strain on food security.

- **Livestock and Fisheries**: Many rural families rely on livestock or fisheries for income. Disasters, particularly floods and droughts, can decimate these industries.
- For example, the 2004 tsunami wiped out thousands of fishing boats and led to long-term economic hardship in coastal regions of India.

2. Infrastructure Damage

- **Transportation Networks**: Disasters like floods, earthquakes, and cyclones often cause widespread damage to roads, railways, airports, and ports, which are critical for trade and daily economic activities.
- The 2001 Gujarat earthquake, for instance, caused significant damage to transportation infrastructure, disrupting supply chains and leading to a loss in productivity.

- **Power Supply and Telecommunications**: A major disruption in electricity and telecommunication services, caused by floods, storms, or earthquakes, can severely hamper industrial production, business operations, and the provision of essential services.
- The 2013 Uttarakhand floods, for example, led to widespread power outages and communication breakdowns in affected areas.

- **Urban Infrastructure**: Cities that are affected by disasters often face substantial damage to residential and commercial properties. The cost of rebuilding homes, schools, hospitals, and businesses can be immense, stretching public finances and slowing down economic recovery.

3. Economic Losses in the Industrial and Business Sectors

- **Manufacturing and Production:** Manufacturing plants and factories are often affected by disasters. Cyclones, floods, or even heatwaves can halt production in key industrial zones.
- For example, the 2015 Chennai floods caused shutdowns in several automotive manufacturing plants and IT hubs, resulting in a loss of industrial output and export revenues.
- **Tourism:** India's tourism sector, which is a major contributor to the economy, is particularly vulnerable to natural disasters. Cyclones, floods, or even pandemics (like the COVID-19 pandemic) lead to a sharp decline in tourism-related revenues.
- For instance, the 2004 tsunami caused massive destruction to coastal tourist destinations in Tamil Nadu and the Andaman and Nicobar Islands.

4. Loss of Employment and Income

- **Direct Impact on Employment:** Natural disasters often result in the loss of employment, particularly in sectors that are directly dependent on physical infrastructure or natural resources. Agriculture, fishing, construction, and manufacturing jobs are particularly vulnerable.
- For example, after the floods in Bihar, many migrant workers lost their jobs, and agricultural laborers saw a sharp decline in income.
- **Long-Term Unemployment:** In addition to immediate job losses, the economic dislocation caused by disasters can lead to long-term unemployment. This is especially true for people whose jobs are in industries that require extensive rebuilding, such as construction, tourism, and agriculture.
- **Migration and Economic Impact:** After a disaster, many people migrate in search of work. In rural areas, the destruction of crops and livelihoods forces families to move to urban centers, often creating pressure on urban job markets, public services, and housing.

5. Impact on National and Regional GDP

- **Short-Term Economic Contraction:** In the immediate aftermath of a disaster, a country's GDP often takes a hit due to a reduction in productivity, disruption in services, and damaged infrastructure.
- For instance, the 2018 Kerala floods resulted in an estimated loss of ₹31,000 crore (around $4 billion USD), which had a significant negative effect on Kerala's economy for that year.

- **Regional Disparities:** Disasters may disproportionately affect certain states or regions, exacerbating regional inequalities.
- For example, frequent flooding in states like Assam, West Bengal, and Bihar hinders long-term economic development and regional growth, leading to economic disparities between disaster-prone and relatively stable areas.

6. Government Spending and Fiscal Strain

- **Increased Public Expenditure:** The government has to divert substantial funds toward relief, rehabilitation, and reconstruction efforts following a disaster.
- This can lead to an increase in government expenditure, which may strain public finances, especially if the disaster is widespread or recurrent.

- **Budget Deficits:** In the face of a large-scale disaster, the central and state governments may have to borrow money to cover reconstruction costs, leading to a rise in budget deficits.
- Additionally, long-term recovery efforts can lead to a delay in other planned development projects, further impacting overall economic growth.

7. Agricultural and Environmental Costs

- **Soil Degradation and Loss of Fertility**: In cases of flooding, there is often soil erosion, loss of fertile land, and long-term environmental degradation. This reduces agricultural productivity over time, causing a decrease in rural incomes and adding to food insecurity. Similarly, droughts lead to water scarcity, depleting groundwater levels and affecting agricultural production.

- **Impact on Natural Resources**: Disasters like cyclones or floods can also damage essential natural resources (forests, water bodies), which communities rely on for livelihood and sustenance.
- For example, floods in forested areas can destroy large swaths of timber, medicinal plants, and wildlife, impacting both the local economy and biodiversity.

8. Impact on Foreign Trade

- **Disruptions in Supply Chains**: Natural disasters, particularly those affecting key ports or industrial zones, can disrupt both imports and exports.
- Cyclones and floods in coastal regions, such as the 2013 floods in Odisha, often delay shipments, affecting international trade and leading to a temporary decline in exports.

- **Damage to Export-Producing Areas**: Key export sectors like agriculture (e.g., tea, spices, rice), textiles, and handicrafts are particularly vulnerable to the impact of disasters.
- For example, the 2011 floods in Pakistan and India affected cotton production, leading to a ripple effect on the textile industry, one of India's largest export sectors.

9. Healthcare Costs

- **Increased Healthcare Spending**: Disasters often lead to an increase in health-related issues, including injuries, diseases, and mental health problems. The healthcare sector becomes overwhelmed, and the government must allocate additional resources to treat the affected population.
- The costs of healthcare in post-disaster are substantial, especially in rural or underserved regions.

- **Long-Term Health Impact**: In the aftermath of disasters, people often face long-term health issues due to inadequate sanitation, spread of diseases, and lack of proper nutrition.
- These conditions further strain the economy by increasing healthcare expenditures and reducing productivity.

Political

1. Government Response and Accountability

- **Political Legitimacy**: The efficiency and effectiveness of government responses to disasters directly influence public perceptions of political legitimacy. Delays or mismanagement in providing aid can erode trust in the ruling government, leading to protests or loss of political support. For example, the slow response to the 2013 Uttarakhand floods led to significant political backlash.
- **Public Discontent**: Poor disaster management can lead to dissatisfaction with the government, potentially impacting electoral outcomes. Politicians may face increased scrutiny, with accusations of corruption or misallocation of resources during relief efforts.

2. Policy Shifts and Political Priorities

- **Disaster Management Policies**: Large-scale disasters often prompt changes in disaster management policies and systems. Governments may introduce new disaster preparedness, risk reduction, and resilience-building initiatives. For example, the 2001 Gujarat earthquake led to the establishment of the National Disaster Management Authority (NDMA).
- **Policy Reforms**: Disasters can also influence broader policy shifts in areas like climate change, urban planning, and environmental conservation. Increased frequency of disasters has pushed for greater focus on sustainable development and climate resilience in political discourse.

3. Political Polarization and Exploitation

- **Exploitation of Disasters**: Political parties may exploit disasters for electoral gains, either by highlighting the shortcomings of the ruling party's response or by positioning themselves as the champions of relief efforts. This can lead to political polarization, as parties try to capitalize on the suffering of affected populations.
- **Regional Tensions**: Disasters can exacerbate existing regional tensions, especially when certain areas are perceived to receive unequal attention or aid. For example, during the Kerala floods, there were concerns about unequal aid distribution between different communities, potentially stoking regional grievances.

4. Civil Society and Political Activism

- **Grassroots Movements**: In the wake of disasters, grassroots movements can emerge, sometimes pushing for political change. These movements often highlight issues such as the inadequacy of disaster preparedness, the exploitation of relief funds, or the lack of proper reconstruction policies.

5. Federalism and Center-State Relations

- **Center-State Dynamics**: Disasters can strain relations between the central government and state governments, especially if the response is perceived as inadequate or politically motivated. The allocation of resources and responsibilities for disaster management often leads to disputes over who should take the lead in providing relief and reconstruction.
- **State Leadership**: In some cases, regional political leaders may gain prominence due to their effective handling of disaster relief efforts. For example, the leadership displayed by Odisha's Chief Minister Naveen Patnaik during cyclone Phailin in 2013 enhanced his political standing at the national level.

6. Security and Political Stability

- **Civil Unrest and Political Instability**: Disasters, especially when they are poorly managed, can lead to civil unrest, protests, or even political instability. Public frustration over inadequate relief efforts can trigger protests, strikes, or demonstrations, which can destabilize governments.
- **Security Concerns**: In certain instances, disasters can have security implications, especially in conflict zones or areas with fragile governance structures. For example, in insurgency-prone regions, the sudden influx of aid or the displacement of people can complicate political dynamics and exacerbate tensions.

Table 1: Significant disasters worldwide by type, according to damage, people affected and death toll, 1963–92 (*De and Joshi*, 1998)

Significant Damage	**%**
Earthquakes	10
Tropical cyclones	30
Drought	22
Floods	32
Other disasters	6

People affected	**%**
Earthquakes	4
Tropical cyclones	20
Drought	30
Floods	32
Famine-food shortage	4
Other disasters	7

Number of deaths	**%**
Earthquakes	13
Tropical cyclones	19
Drought	3
Floods	26
Epidemics	17
Landslides	7
Storms	6
Other disasters	9

Economic Impact

- The World Bank, using a standard assessment technique, has estimated that the total damage and losses caused by the earthquake and tsunami amounted to approximately US$4.45 billion, or almost 100 percent of Aceh's GDP in 2003.
- Of this total, 60 percent is from damage 40 percent is from losses of income owed to the economy.
- Around 78 percent of the total damage and losses were borne by the public sector.
- Around one- third of the road networks, schools were destroyed by the tsunami.

Social Impact

- The Indian Ocean Tsunami killed an estimation of 230,000 lives and left 1.7 million people shelter less.
- In northern Sumatra, the epicentre of the event, 1500 villages were destroyed.
- More than 5 million people were injured and needed emergency relief, food and water.
- In addition to the immediate effects there was also the risk of illness from mixing fresh, sewage and saltwater.
- After the tsunami, the rates for cholera, hepatitis and other problems increased.

Psychological effects

- According to the Conservation Resource Model, people try to protect resources such as objects (housing, possessions, etc.), social roles (employment, marriages, etc.), energy (time and monetary investments), and personal characteristics (e.g., self-confidence).
- The threatened or actual loss of these resources as caused by a natural disaster leads to psychological distress.
- Frequently observed conditions such as minor emotional disorders seldom come to the attention of psychiatrists but may negatively affect social relationships and work performance.
- Commonly detected symptoms are fatigue, tics, and cognitive experiences such as confusion, impaired concentration, and attention deficit disorder.
- Emotional signs such as anxiety, depression, and grief, as well as behavioral effects such as sleep and appetite changes and substance abuse, were also reported.

- Even effects on suicide rates were detected: earthquake victims (people who had lost family members residing with them, were injured, or experienced property loss) were 1.46 times more likely than non-victims to commit suicide.
- All these effects can be mild and transitory or can lead to Post Traumatic Stress Disorder (PTSD).
- The mental states of victims can include three stages: an immediate reaction involving distressing symptoms accompanying adaptive stress; the post-immediate phase, which includes symptoms of maladaptive stress (confusion, agitation, and occasionally neurotic or psychotic reactions); the long-term sequel phase, which involves a return to normal health or the onset of PTSD, which can sometimes yield a chronic phase that involves personality changes.

- These surveys make it possible to monitor the most fragile segments of the population, including people with pre-existing mental illness, racial and ethnic minorities, and children, in which symptoms may differ depending on age.
- Gender differences arise as well: for instance, after an earthquake, women report greater emotional distress and mental health problems than do men, but the occurrence of addiction disorders among women is much lower.

Ecology

- In recent years, there has been a steep increase in the frequency and intensity of environmental disasters across the world. These disasters pose a great threat to not only human existence but also complex ecosystems.
- Cases of forest fires, heat waves, extreme winters, erratic rainfall, prolonged droughts have been increasing every year.
- India with its diverse geography and fragile ecosystems has been facing the brunt of environmental disasters too.
- According to a report titled India 2022: An Assessment of Extreme Weather Events published by the Centre for Science and Environment (CSE) in 2022, India recorded extreme weather events on 241 of the 273 days between January 2022 to September 2022.
- This accounts for 88 percent of this nine month period wherein there was an extreme weather event occurring across the country. The report estimates that these events have claimed 2,755 lives, affected 1.8 million hectares of crop area, destroyed over 400,000 houses and killed close to 70,000 livestock.

- The North East Indian Region has been particularly vulnerable to the impacts of such disasters.
- According to this report by CSE, Assam reported the highest number of damaged houses and animal deaths and the North East Indian Region is estimated to have accounted for 783 human lives lost as a result of these environmental disasters/extreme events.
- Further, the report also suggests that in 2022, Eastern and northeastern India saw its warmest and driest July in 121 years.
- The ecological richness of the North East Indian Region is unparalleled. From the snowlines of Arunachal Pradesh and Sikkim, to the rainforests of Meghalaya and Assam, the region sheltered an array of unique flora and fauna living in dense natural forests.
- However, this ecological diversity also renders the region susceptible to the disruptive effects of environmental disasters. Take, for instance, the devastating floods that have become an almost annual occurrence in the region.
- The Brahmaputra River, a lifeline for millions, often transforms into a ruthless force of nature during the monsoon season.

- The resulting floods can wreak havoc on both human settlements and ecosystems, leading to soil erosion, sediment deposition, and alteration of river channels.
- Such alterations disrupt the delicate balance of ecosystems, displacing species and compromising biodiversity.

- Forest fires, another recurrent environmental disaster, have been plaguing the North East Indian region as well.
- The 2019 fires in the Dzukou Valley on the Manipur-Nagaland border highlighted the dangers posed by changing weather patterns and human activities like slash-and-burn agriculture.
- The fires not only devastated vast tracts of forest cover but also released massive amounts of carbon dioxide into the atmosphere, exacerbating climate change.

- Lastly, the importance of raising awareness and fostering a sense of ownership among local communities cannot be overstated.
- The indigenous communities of the North East have coexisted with their ecosystems for generations, often possessing knowledge that is crucial for maintaining ecological balance.
- Rampant mining and deforestation for extracting construction materials has made the mountain ecosystems fragile resulting in frequent cases of landslides across the region.
- This year we have seen unprecedented rainfalls causing massive landslides in Nagaland, Meghalaya and part of Assam. During the monsoon, such events become deadly and affect thousands of people by either displacing them or taking away their means of livelihood.
- Soil erosion has threatened the existence of Majuli, world's largest riverine island located in Assam. Every year during the annual floods, large tracts of land get washed away and people lose their lands and livelihoods.
- Building resilient ecosystems requires a multi-pronged approach that integrates scientific research, community involvement, and policy interventions.

- The disastrous landslides in the Darjeeling region in 2017, triggered by heavy rainfall and rampant deforestation, demonstrated the importance of understanding local ecological contexts.
- In this case, the landslides could have been prevented or mitigated through better land-use planning and reforestation efforts.
- Furthermore, the North East Indian region underscores the transboundary nature of environmental disasters.
- The region shares borders with Bangladesh, Bhutan, Myanmar, and China, making it imperative for nations to collaborate in disaster management and ecosystem preservation.
- Climate change, deforestation, and pollution know no borders, and the effectiveness of any intervention is limited if not coordinated on a regional scale.
- Platforms for cooperation and information-sharing are essential in ensuring the holistic health of ecosystems in the face of disasters.

- Disaster preparedness, ecosystem restoration, transboundary cooperation, and community involvement emerge as crucial pillars in building and sustaining resilient ecosystems.

Health impacts of disasters

- The World Health Organization (WHO) defines health not merely as an absence of disease, but as a state of physical, mental and social well-being.
- A potential omission of this definition is 'the ability to adapt and self-manage in the face of social, physical, and emotional challenge's.
- Disasters are an extreme source of such challenges, and health risk exposure, with notable profound effects on human physical, mental and social health, as well as economic and environmental effects, and access to healthcare services.

- Natural Disaster, hazard-specific health effects such as increased fractures and crush injuries in earthquakes, increased drownings in floods and tsunamis, increased mortality rates during extremes of temperatures and increased suffocation in landslides and avalanches.
- The disaster literature demonstrates an association between many symptoms and clinical conditions with exposure to a disaster.
- Researching this association immediately after disasters is challenging.
- Issues include the unpredictable nature of disasters requiring rapid development of ethical, relevant research protocols; access and safety in chaotic, dangerous environments; inclusion of vulnerable potentially traumatised populations; and potential impedance of rescue activities.

- Randomised controlled trials are ethically and practically difficult in the disaster environment.
- Over the last 20 years, increasing numbers of longer-term prospective cohort studies have provided stronger evidence of an increase in prevalence, incidence and association of many health conditions and symptoms following disasters, over immediate, intermediate and long-term periods, along with new incident conditions, and deterioration of pre-existing, chronic conditions.
- Case studies have contributed by describing new conditions such as tsunami lung and tsunami sinusitis.

Communicable, or infectious, health effects

- In well-resourced countries, infectious outbreaks are uncommon following disasters, except where the primary disaster is infectious itself (e.g., the global SARS-Co V-2 (Severe Acute Respiratory Syndrome-associated Coronavirus 2).
- Even with significant disruption to utilities, studies from Hurricane Katrina and Hurricane Sandy showed no significant increase in infectious disease outbreaks.
- However, in lower-resourced countries and in humanitarian disasters, infectious diseases are a major concern, predominantly due to increases or changes in locally endemic organisms including acute respiratory infections and gastroenteritis.
- Disruption of water and sanitation, displacement of populations, crowding in evacuation centres, poor pre-event health status and disruption of health services, may all impact on the risk of water, food or air- home infectious disease.
- In poorly vaccinated populations, vaccination programs for highly infectious and deadly diseases such as measles and tetanus, need to be urgently established.

- Infectious disease tends to present a different time course to acute injury in disasters.
- Following the Great East Japan Earthquake Tsunami (GEJET) a significant increase in infectious disease hospitalisations occurred later, at two weeks, and was predominantly pneumonia due to usual community-acquired pathogens.
- Mortality attributed to pneumonia was significantly increased in the first 12 weeks peaking in the second week.
- Half the cases of pneumonia were contracted whilst in evacuation centres.

Non-communicable, or non-infectious, health effects

- The scientific literature reports disaster health effects on most body systems, including cardiac, respiratory, endocrine, dermatologic, musculoskeletal, gastrointestinal, ocular, otologic and neurological effects.
- This section aims to provide a broad overview of non-communicable health effects with discussion of new incident disease and symptoms, including Medically Unexplained Physical Symptoms (MUPS), and of chronic disease.
- To fully realise the physical health impact of disasters it is crucial to understand the temporal nature of these effects.
- That said, we are making progress in developing insights on the temporality of physical health impact, although we are not yet at the point where we can reliably plot the prevalence or risk for particular health problems over the course of weeks, months and years in detail.

Cardiac effects

- An increase in the incidence of acute cardiac events such as hypertension, acute myocardial infarction (AMI), cardiomyopathy and non-cardiac chest pain (NCCP) following disasters is increasingly well-documented with early evidence of an influence on heart failure (HF) and ventricular tachyarrhythmia.
- Hypertension was considered one of the greatest burdens to healthcare in the early stages of the GEJET and Hurricane Katrina.
- A high background prevalence of pre-existing hypertension in these populations was compounded by an increase in blood pressure (BP) in those with no history of hypertension (92%), and in those with pre-existing hypertension, both in those taking their usual medications (10%), and those not taking them.
- A significant increase in AMIs was associated with the 2010 Christchurch earthquake, occurring during the first two weeks post-event [18]. The WTC attack was associated with a significant increase in cardiovascular admissions of lower Manhattan residents to NYC hospitals within the first five weeks.

- During the second week there was a 72.7% relative increase in cardiovascular admissions compared with 14.7% in the control area.
- Post Hurricane Katrina an increased incidence of AMI was sustained at two years and three years with a three-fold increase observed over both periods.
- Linking back to mental health, an association has been reported between PTSD and vascular con-ditions following disasters.

Psychosocial impact due to disaster

Introduction:

- The word disaster either natural or man-made reminds us the most horrific images of human sufferings in multifarious ways. However, until recently there has been a general tendency to consider the basic needs of the affected people and therefore, the emphasis was on providing curative care, food, shelter, relief, immunization, income generating activities, and others.
- Addressing mental health and psychosocial needs has often been considered as secondary and accessory to the basic needs.
- Nevertheless, a global realization has evolved and emerged that the loss experience and emotions attached to it are complex and is much more than just the superficial aspect reported by media as witnessing disasters through the eyes of television camera hardly gives true insight into the psychological suffering (Boer and Dubouloz, 2000).
- As the deep hurt and anguish caused by the loss of human life, as well as the disruption of daily life are far more difficult to overcome, more so in man-made disasters, like riots. Therefore, emotional needs have to be given priority along with relief, rehabilitation and care of physical health.

- Thus, mental health management as an essential component in all disaster management plans either for man-made or for natural has recently received top attention from the Department of Mental Health of the World Health Organization (WHO), which defines "health" as a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity.
- Mental health is an important component of the well being of people exposed to disasters and critical to successful rehabilitation and development of the societies in the aftermath.
- Many immediate reactions largely depend upon the physical explicitness of a disaster, such as, exposure to extreme danger, witnessing near one's deaths, helplessness, hopelessness, and trauma of having to choose between one's own survival and that of others (Murthy, 2000).
- Therefore, mental health at individual and community level becomes necessity to be maintained to prevent possible psychiatric disorders after any disaster and to maximize the progress of successful rehabilitation and development of communities in post disaster scenario.

SIGNIFICANCE OF DISASTER PSYCHOSOCIAL CARE

- Psycho-social support in the context of disasters refers to comprehensive interventions aimed to address a wide range of psycho-social problems arising in the aftermath of a disaster.
- For example, listening actively to the survivors, helping them to restart their daily routine works, helping them to get connected with their near and dear ones, providing legal and paralegal help, may also be considered as parts of this intervention package.
- Any intervention that helps in reducing the level of actual and perceived stress stemming out of the disastrous situation, and in preventing adverse psychological and social consequences among disaster affected people can be defined as disaster psycho-social support/care services.
- These services can be provided by imparting basic skills training on the subject to the community level people/workers (CLWs). He/she can be a teacher, anganwadi worker, panchayat member, social workers, local NGO people, local youths, members of mahila mandals, etc.

- Disaster mental health services refer to the medical interventions for identification and management of manifest stress related clinical psychological signs/symptoms or of the mental disorders among disaster affected persons.
- In addition, interventions aimed at mental health and psychological well-being promotion, and prevention of psychological/psychiatric symptoms among disaster affected population are also included under disaster mental health services.
- These services need expertise services of psychiatrist, psychologists, mental health professionals, psychotherapist, etc.

The key goals of adopting a community model of disaster mental health and psycho-social care are:

- Preventing long-term psychiatric disorders in the disaster affected society;
- Providing relief from mental suffering and psychological distress;
- Maintaining mental well-being and equilibrium;
- Promoting positive mental health;

- All the above four become more important for the optimal utilization of resources and economic opportunities offered through the community rehabilitation and development programs (WHO, Division of Mental Health, 1991);
- Strengthening the social support networks in the affected area;
- Facilitating the community participation in all activities taken place in mitigation and relief and rehabilitation phase.

COMMON PSYCHOLOGICAL REACTIONS FOLLOWING A DISASTER

- First of all one should understand that any psychological and emotional reaction is not itself all negative, for it can increase the chances of the survival of the victim.
- Stress becomes a threat to mental health when it overwhelms the capacity of the victims to cope with the new situations by mastering their own reactions. A cauldron of emotional reactions can come to boil after a disaster. Although people react differently to traumatic events on the basis of their experiences and personality, there are number of common responses that are experienced by the majority of those involved. These common post-disaster reactions include
- **emotional** (panic attacks, shock, fear, anger, sadness and guilt feeling),
- **psychosomatic** (sleep disturbances, physical problems like muscle tension, palpitation, headaches, nausea, diarrhea or constipation, and breathing difficulties),
- **cognitive** (repeated thoughts and involuntarily triggering the memories, nightmares, confusion, flashbacks, difficulty in concentrating and making decisions, memory problems, shortened attention span, etc),

- **behavioural and attitudinal** (disruptions in social relationship, poor motivation and concentration, lethargy, hopelessness, etc) **difficulties**.
- Normally, these reactions ‘settle’ over the first week.
- If, however, they remain protracted and intense and moreover, If symptoms persist for a period of three months or after that the person is very likely to suffer from various psychological disorders.

- When one talks of common reactions among the disaster affected population, one should also differentiate between the psycho-social reactions exhibited/expressed by adults and children. Children are affected by any disaster just the way adult are, although their reactions might differ from those of the adults.
- Disasters disrupt the sense of well-being by destroying normal predictable and consistent life routines of children thus, deeply hamper the process of healthy psychological and personality development.
- Since children have limited capacities to process information their sense of what happened is often not realistic and they are not able to comprehend the totality of the situation. Therefore, before planning for any type of psychosocial interventions, one needs to understand the common psycho-social reactions shown by children in various age groups.

- The psycho-social aspects of disasters on human beings have been acknowledged as an international agenda (WHO, 1992). However, in India, the psycho-social aspects have never been emphasized until very recently after tsunami, 2004. The Bhopal gas tragedy (1984) was the most important disaster to draw the national attention due to its severe impact and the sensitivity of the politico-economic issues involved. The psycho-social impact was studied systematically although intervention programmes were more of psychiatric in nature. Latur earthquake (1993), and Andhra Pradesh Super Cyclone (1996) were disasters in which mental health professionals took an active part in terms of providing mental health services and undertaking research to study the psycho-social impact of these disasters.

- The ICMR studies over last twenty years have provided strong base for integration of mental health services with general health care services and sensitization of the community members and rescue workers. However, it has been difficult to integrate the mental health services at micro and macro level. Recently, National Disaster Management Agency and National Institute of Disaster Management has mixed experiences in providing health care services in disaster situations. However, the finer details of the mechanisms and strategies for integration of mental health services with general health care services still need to be worked out.

- In the post Tsunami phase in India, the WHO along with the Department of Social Welfare, United Nations Team for (UNTRS), and partners have developed a model for providing sustained, low cost community-based volunteer provided support systems. Community level workers who are the anchor for this programme are selected from various categories of people, including teachers, health workers, and members of self help groups etc, who have volunteered for this purpose. A cascading system of training was developed and in Tamil Nadu, 2813 Community Level Workers (CLWs) were trained in the 11 affected Districts. They were able to support more than 30,000 families and 150,000 individuals.

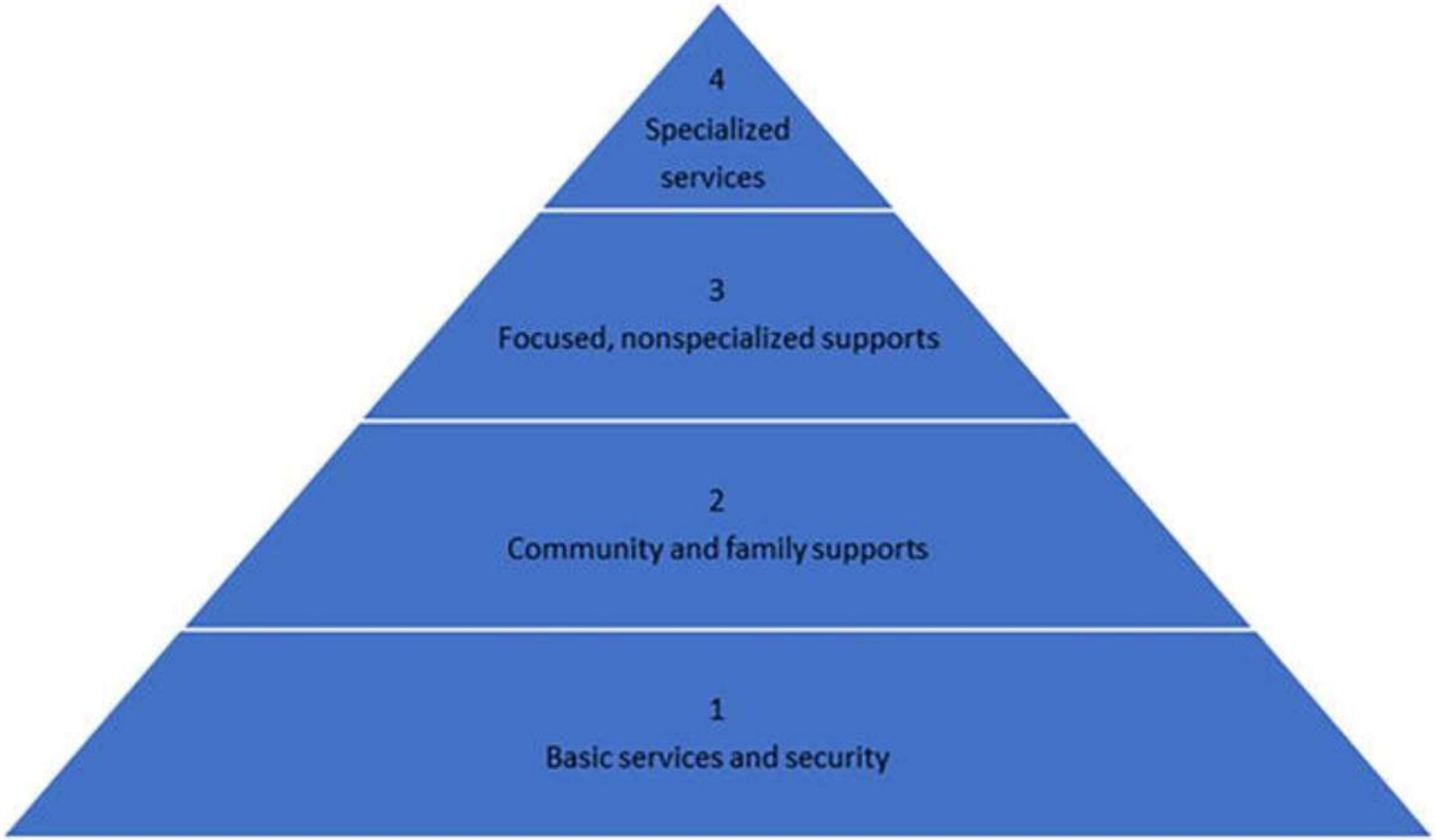

Figure 3. Intervention pyramid for mental health and psychosocial support in emergencies.

ROLE OF PSYCHOSOCIAL CAREGIVER

- Any community when faced with a disaster responds in its own way to the situation. One such response is the reaction of several local people or groups of people who immediately come forward to help in one way or the other in order to alleviate the situation. These people are known as the ***community level workers* (CLWs).** They play a crucial role in the relief and rehabilitation efforts after the community has experienced a disaster.
- These CLWs mostly are people who belong to the community and are likely to know the area well and have close ties with several people in the locality. This enables them to work in a sustained and intense manner with the community.
- These CLWs might include

Anganwadi functionarie
Auxiliary Nurse Midwives (ANMs)

Schoolteachers
NGO/CBO workers
Volunteers
Local community leaders
Temple priests
Panchayat members
Revenue Inspectors
Women's group members

- Generally these CLWs do have a helping tendency, but they often don't know whether they are doing it right or wrong. ***The CLWs are a vital link between the disaster survivors and several helping agencies (NGOs, Governmental agencies) that come from outside the community.*** They provide help to the survivors from the initial phase, immediately after the disaster has occurred. Psychological problems following disaster often tend to be neglected, as they are relatively less visible when compared to the damage to life, physical health and property. It's important to remember that psychological problems often occur and if unattended to, would hinder the functioning of the survivor. Hence, early identification of this problem, followed by intervention, helps the survi vor to recover.

- A psychosocial caregiver from the community plays a critical role in rebuilding efforts after any community has experienced a disaster. In the Orissa cyclone disaster the 'Snehkarmis' and in the Gujarat earthquake the "Viklang bandhus' provided psychosocial care to the survivors of the disasters.
- Under normal circumstances most people can take care of their problems. Tsunami being an abnormal situation, people's equilibrium has been temporarily upset because of the emotional reactions they are now experiencing. Till the time they can develop successful coping strategies and handle their lives competently, they need emotional support.
- It is like extending help to a person who has fallen on the ground. Extend your hand to help the person sit, and then slowly make him or her stand. Then walk a few paces with him / her and gradually he or she will walk independently. This diagram would help in understanding at a glance the three areas of focus for psychosocial interventions.
- **Relief from stress, ability to talk about the experience and passage of time usually leads to the re-establishment of equilibrium. Individuals find comfort and reassurance when told that their reactions are normal and understandable in every way. Public information about normal reactions, education about ways to handle them, and early at- tention to problematic symptoms can hasten recovery and prevent long-term problems.**

Demographic aspects: Gender

- ***Disaster impacts are not gender-neutral:***
- The impact of disaster caused by natural hazards is not gender-neutral.
- A 20-year study on gendered nature of natural disaster shows that in societies where the socioeconomic status of women is low, natural disasters kill more women than men—both directly and indirectly through related post disaster events.
- The study also shows that disasters kill women at a younger age than men, and that the impacts of natural disasters are never merely determined by nature on its own.
- The reason for the difference in mortality lies in the fact that women in general have lower socioeconomic status which, in turn, leads to unequal access to opportunities and unequal exposure to risks, making them more vulnerable to natural disasters.

- Women's increased vulnerability to disaster has many dimensions. Women's reproductive roles can limit their chances of survival—those in their final stages of pregnancy and women with young kids are less mobile. If basic health care infrastructure is severely damaged and access to obstetrical care is limited, chances of miscarriage as well as maternal and infant mortality increase.
- Social norms and gendered roles may significantly affect women and girls' ability to survive the impacts of a disaster.
- Studies on the impact of the 2004 tsunami highlight that differences in self-rescue ability are partly determined by learned skills such as swimming and tree climbing which, in many contexts, are primarily taught to and practiced by men and boys. Further, dress codes can restrict women's ability to move quickly in times of disaster or restrict them from learning survival skills.
- Social norms may prevent women from relocating without the consent of a male family member, restricting women from getting information about evacuation centers or how to escape during disasters.

- Women also have limited access to economic resources as well as information and technology, increasing their vulnerability and adaptive capacity to disaster.
- Traditional expectations and home-based responsibilities limit women's mobility and opportunities for political involvement, education, access to markets, and a myriad of other resources—the lack of which reinforces the cycle of vulnerability.
- Inheritance laws and traditions, marriage arrangements, banking systems, and social patterns that reinforce women's dependence on fathers, husbands, and sons all contribute to their unfavourable access to resources and their lack of power to change things.
- A key gendered impact of disaster is women's increased post-disaster workload. Women are often in the frontline of reestablishing households in relocated or reconstructed sites and collecting basic household needs such as water, food, hygiene items, and fuel.
- Such engagements may cause girls to drop out of school, disrupt women's skill building and other livelihood opportunities, and negatively impact women and girls' time poverty.

Existing gender roles can change in disaster settings

- Disasters disrupt commerce and markets and destroy productive resources and infrastructure.
- In some cases, women become breadwinners as the male members are killed by disaster or choose to stay with the remaining property in order to safeguard the little that remains.
- This may put additional burden on women to find new sources of livelihood and continue the traditional role of taking care of the family.

Gendered response can reduce long-term social consequences

- The immediate impact of disasters—displacement; destruction of homes, schools, and health services; loss of financial security and livelihood; and disruption of existing social protection mechanisms—may further expose women and girls to devastating, long-term effects and consequences.
- These may include early and forced marriages, trauma, increased vulnerability to gender-based violence, exposure to sexually transmitted infections and HIV, and unintended pregnancies.
- Failure to apply gender lens in disaster risk management and response can reinforce or perpetuate these social consequences on women and girls.

Women as agents of change for disaster risk reduction and strengthening resilience

- Engaging and empowering women are beneficial means of strengthening resilience to disaster risks.
- Just as gender roles and relations shape vulnerability to disaster, conversely they also shape people's capacity to prepare, withstand, and recover.
- Women have particular experience and skills to contribute to disaster risk management due to their role as primary caregivers within the family and often within the wider community.
- They have strong local knowledge of natural resources, surroundings, and valuable links with others in the area.
- If women's specific perspectives are not factored into disaster risk management and response, the knowledge is lost.

Focus group discussions with women living in relief camps. Pakistan.

Climate change and Urban disorders

Climate change significantly impacts urban areas, leading to various environmental, social, and health-related challenges collectively termed as "urban disorders." These disorders encompass a range of issues, including increased heatwaves, flooding, air pollution, and the spread of vector-borne diseases, all of which disproportionately affect urban populations.

Urban Heat Islands (UHIs)

- Urban areas often experience higher temperatures than their rural surroundings, a phenomenon known as the Urban Heat Island effect. This is primarily due to heat-absorbing materials like concrete and asphalt, reduced vegetation, and waste heat from buildings and vehicles.
- UHIs exacerbate heatwaves, leading to increased energy consumption for cooling and heightened health risks such as heatstroke and respiratory problems. For instance, studies have shown that summer daytime temperatures can be up to 10°C higher in city centers compared to rural areas.

Increased Flooding and Sea-Level Rise

- Climate change contributes to rising sea levels and an increase in extreme weather events, leading to more frequent and severe urban flooding. Coastal cities are particularly vulnerable, facing threats to infrastructure, housing, and public health.
- The expansion of urban areas into floodplains and inadequate drainage systems further exacerbate the risk. The devastating floods in Valencia, Spain, in October 2024, which resulted in over 200 fatalities, underscore the urgent need for urban planning that incorporates climate risk assessments.

Air Pollution and Health Impacts

- Urbanization and climate change have led to deteriorating air quality in cities, with increased emissions from transportation, industry, and energy production.
- Poor air quality is linked to respiratory and cardiovascular diseases, and climate change can exacerbate these effects by increasing the concentration of pollutants.
- Additionally, higher temperatures can lengthen the pollen season, intensifying allergies and asthma cases among urban residents.

Spread of Vector-Borne Diseases

- Warmer temperatures and altered precipitation patterns due to climate change expand the habitats of disease-carrying vectors like mosquitoes.
- Urban areas, with their dense populations and often inadequate sanitation, become hotspots for outbreaks of diseases such as dengue and malaria.
- In 2024, Bangladesh experienced its worst dengue outbreak, with over 400 deaths and more than 78,000 hospitalizations, highlighting the vulnerability of urban populations to vector-borne diseases in a changing climate.

Mental Health and Ecological Grief

- The impacts of climate change on urban environments also extend to mental health. The degradation of urban ecosystems, loss of green spaces, and increased frequency of natural disasters contribute to feelings of anxiety, depression, and ecological grief among city dwellers.
- The psychological stress of living in rapidly changing urban landscapes underscores the need for integrating mental health considerations into urban planning and climate adaptation strategies.

Mitigation and Adaptation Strategies

Addressing the challenges posed by climate change in urban areas requires a multifaceted approach:

1. Enhancing Green Infrastructure: Planting trees, creating parks, and developing green roofs can mitigate UHIs, improve air quality, and provide recreational spaces that enhance mental well-being. For example, urban vegetation can reduce ambient temperatures through shading and evapotranspiration, offering a natural cooling effect.

2. Improving Water Management: Upgrading urban drainage systems, restoring natural waterways, and implementing sustainable urban drainage systems (SUDS) can reduce flood risks. Cities like Los Angeles are learning from Australia's experience with wildfires, emphasizing the importance of robust water infrastructure to combat fire-related urban disorders

3. Strengthening Public Health Systems: Enhancing disease surveillance, improving healthcare infrastructure, and promoting public awareness are crucial to combat the spread of climate-sensitive diseases. Integrating climate and health data can help cities develop proactive strategies to address the climate-health crisis.

4. Promoting Sustainable Urban Planning: Incorporating climate resilience into urban development, enforcing building codes that consider future climate scenarios, and avoiding construction in high-risk areas are essential steps. The recent wildfires in Los Angeles highlight the consequences of building in fire-prone areas without adequate planning and preparedness.